

Gradient Descent

Linear Regression (Recap)

- Goal:
 - Given a training set, learn θ
 - Such that $h_{\theta}(x)$ is as close as possible to y

Least ~~Mean~~ Squares (LMS)

- For a dataset with n samples, minimize

$$J(\theta) = \frac{1}{2} \sum_{k=1}^n (h_{\theta}(x^{(k)}) - y^{(k)})^2$$

Cost Function/Loss Function

$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

Linear Regression

- Goal: Choose θ to minimize $J(\theta)$

$$\theta^* = \operatorname{argmin}_{\theta} J(\theta)$$

- How?
- Gradient Descent

Gradient Descent

Cost Function

$$J(\theta) = \frac{1}{2} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

- Start from an initial θ
- Repeatedly update θ as follows

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

(j=0 to d) θ_j α $\frac{\partial}{\partial \theta_j} J(\theta)$ $\rightarrow$ gradient value

constant

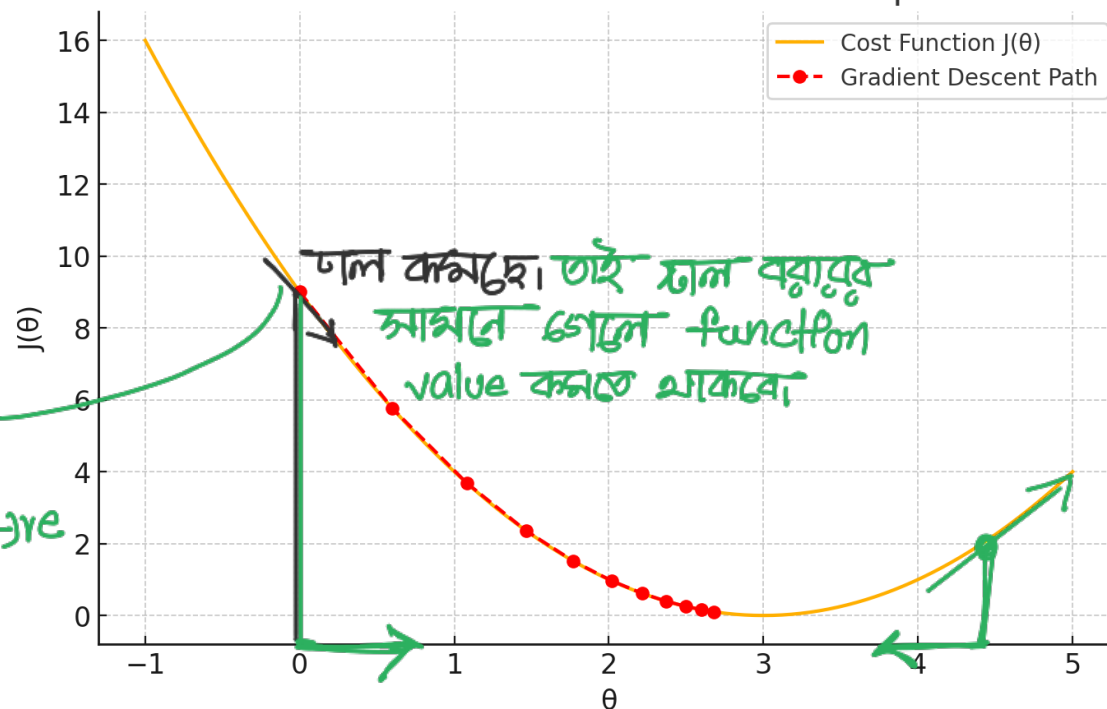
- θ_j corresponds to the coefficient of the x_j

- α is called learning rate

✓ Constant

✓ Decaying

Gradient Descent Visualization with Respect to θ



Gradient Descent

For linear regression, how we perform gradient descent

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

$$\begin{aligned} \frac{\partial J}{\partial \theta_j} &= \frac{1}{2} \sum_{k=1}^n \frac{\partial}{\partial \theta_j} (h_{\theta}(x^{(k)}) - y^{(k)})^2 \\ &= \frac{1}{2} \sum_{k=1}^n 2(h_{\theta}(x^{(k)}) - y^{(k)}) \frac{\partial}{\partial \theta_j} h_{\theta}(x^{(k)}) \\ &= \sum_{k=1}^n (h_{\theta}(x^{(k)}) - y^{(k)}) \frac{\partial}{\partial \theta_j} \left(\sum_{i=0}^d \theta_i x_i^{(k)} \right) \\ &= \sum_{k=1}^n (h_{\theta}(x^{(k)}) - y^{(k)}) x_j^{(k)} \end{aligned}$$

Cost Function

$$J(\theta) = \frac{1}{2} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\begin{aligned} h_{\theta}(x) &= \theta^T x \\ &= \sum_{i=0}^d \theta_i x_i \end{aligned}$$

Gradient Descent

$$J(\theta) = \frac{1}{2} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Cost Function

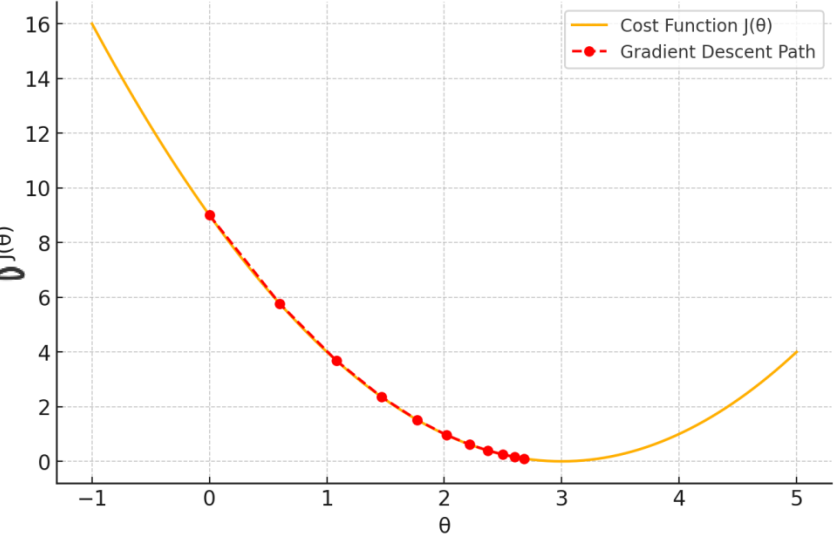
- Start from an initial θ

Repeat until convergence

$$\theta_j := \theta_j + \alpha \sum_{k=1}^n (y^{(k)} - h_{\theta}(x^{(k)})) x_j^{(k)}, \quad (\text{for every } j)$$

नहीं 25M this is still costly, repeatedly 2602

Gradient Descent Visualization with Respect to θ



Reference

Convergence

Gradient = 0 or almost 0 means converge হবে, converge is cost function tends to 0 but it's infinite loop হয়, That's why we wait for gradient = 0 or almost 0.

Cost function convex নাও হতে পারে অনেকগুলো Local অথবা global minima থাকতে পারে।